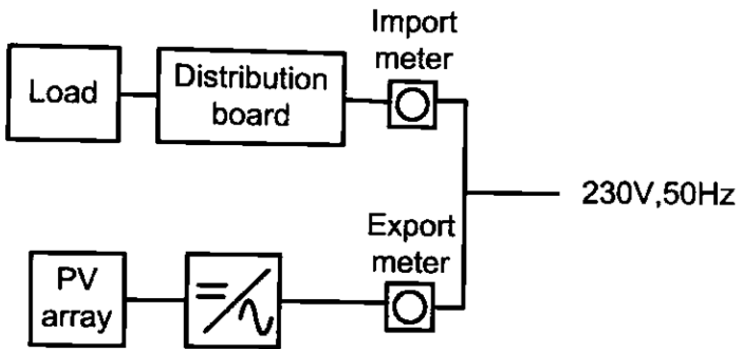
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Renewable Generation Technologies

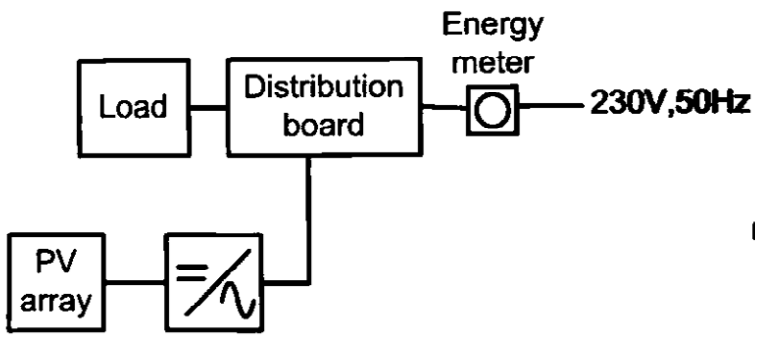
Grid-Connected PV System Design

Introduction

- There are two types of connection: Direct feed-in and indirect feed-in.
- In direct feed-in method, power consumption and power generation are segregated and measured independently by import and export meters.
- Indirect feed-in transfers power from PV to the grid via distribution board. This method is adopted by Net Energy Metering (NEM) and Self Consumption schemes.

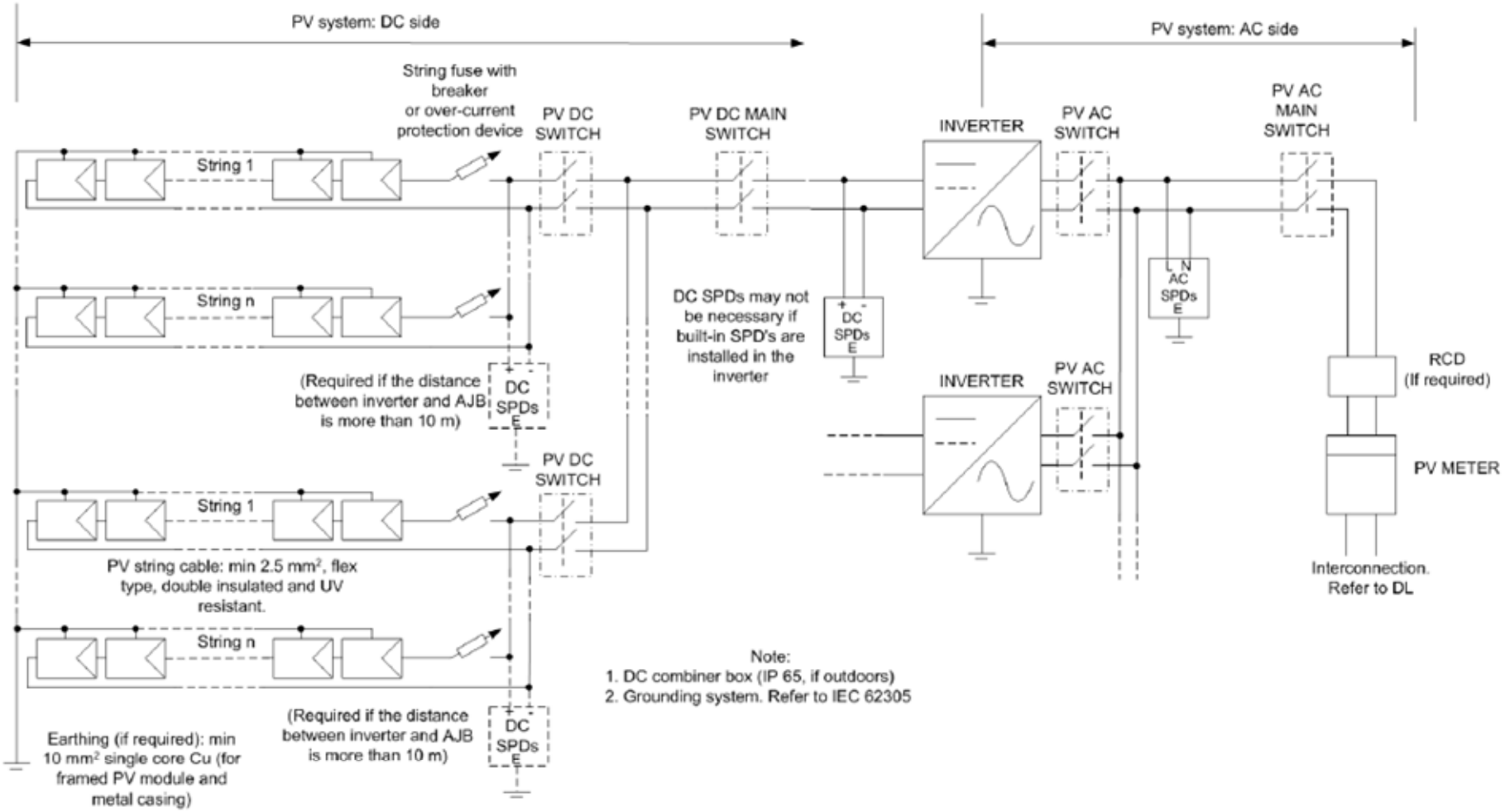


Direct feed-in

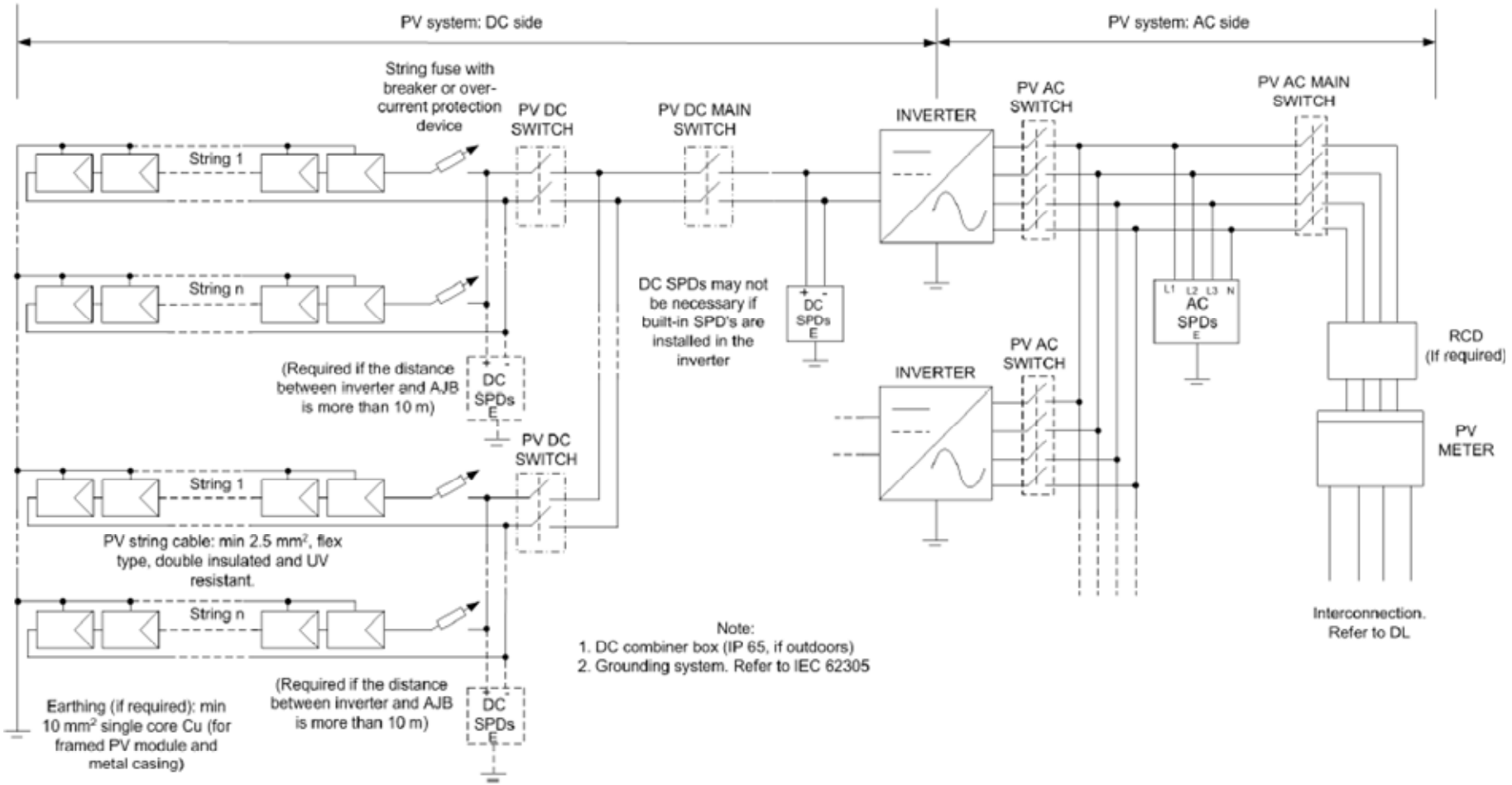


Indirect feed-in

Single-Phase System



Three-Phase System



Sizing of PV System

- The philosophy of design is fundamentally balancing the requirements, constraints, resources and components.
- Three main factors to be considered for rooftop solar photovoltaic (PV) system design are:
  - Energy constraint
  - Space constraint
  - Budget constraint



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Energy Constraint

Design Procedures

- The number of PV modules required are determined based on the energy requirement.
- The following steps that involved are:
  1. Determine the amount of energy to be supplied by the PV.
  2. Determine the minimum power of PV array required.
  3. Determine the minimum number of PV modules required.

ENERGY CONSTRAINT

Required Energy

- The energy per annum required to be generated by the PV arrays is

$$E_{req} = \frac{\varepsilon}{100\%} \times \frac{12}{n} \times \sum_{n=1}^{12} E_m$$

E_{req} = energy required per annum (kWh)
ε = percentage of energy to be supplied using PV (%)
E_m= energy consumption for month m (kWh)
n = number of months with available data

ENERGY CONSTRAINT

PV Array Power

- Based on the annual energy required by the PV array, the minimum power of PV array is calculated as

$$P_{arr_stc} = \frac{E_{req}}{PSH \times n_{inv} \times n_{cab} \times k_{dir} \times k_{mm} \times k_{tem_p_ave}}$$

P_{arr_stc} = maximum power of PV array at STC (Wp)
PSH = peak sun hour per annum (h)
n_{inv} = maximum efficiency of inverter
n_{cab} = efficiency of DC cable from PV to inverter
k_{dir} = de-rating factor due to dirt
k_{mm} = de-rating factor due to module mis-match
k_{dir} = de-rating factor of power due to average maximum temperature

ENERGY CONSTRAINT

Calculation

- The de-rating factor of power due to average maximum temperature can be calculated as:

$$k_{tem_p_ave} = 1 + \left[\left(\frac{\beta_{power}}{100}\right) \times T_{amb}\right]$$

β_{power} is temperature coefficient of power at STC (% per ℃)
T_{amb} is ambient temperatures(℃)

- The minimum number of PV modules is calculated as :

$$N_{\min_mod} = \left[\frac{P_{arr_stc}}{P_{mod_stc}}\right]_{round\ up}$$

$N_{\min_mod}$ is minimum number of PV modules required
P_{arr_stc} is maximum power rating of PV module at STC (Wp)



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Space Constraint

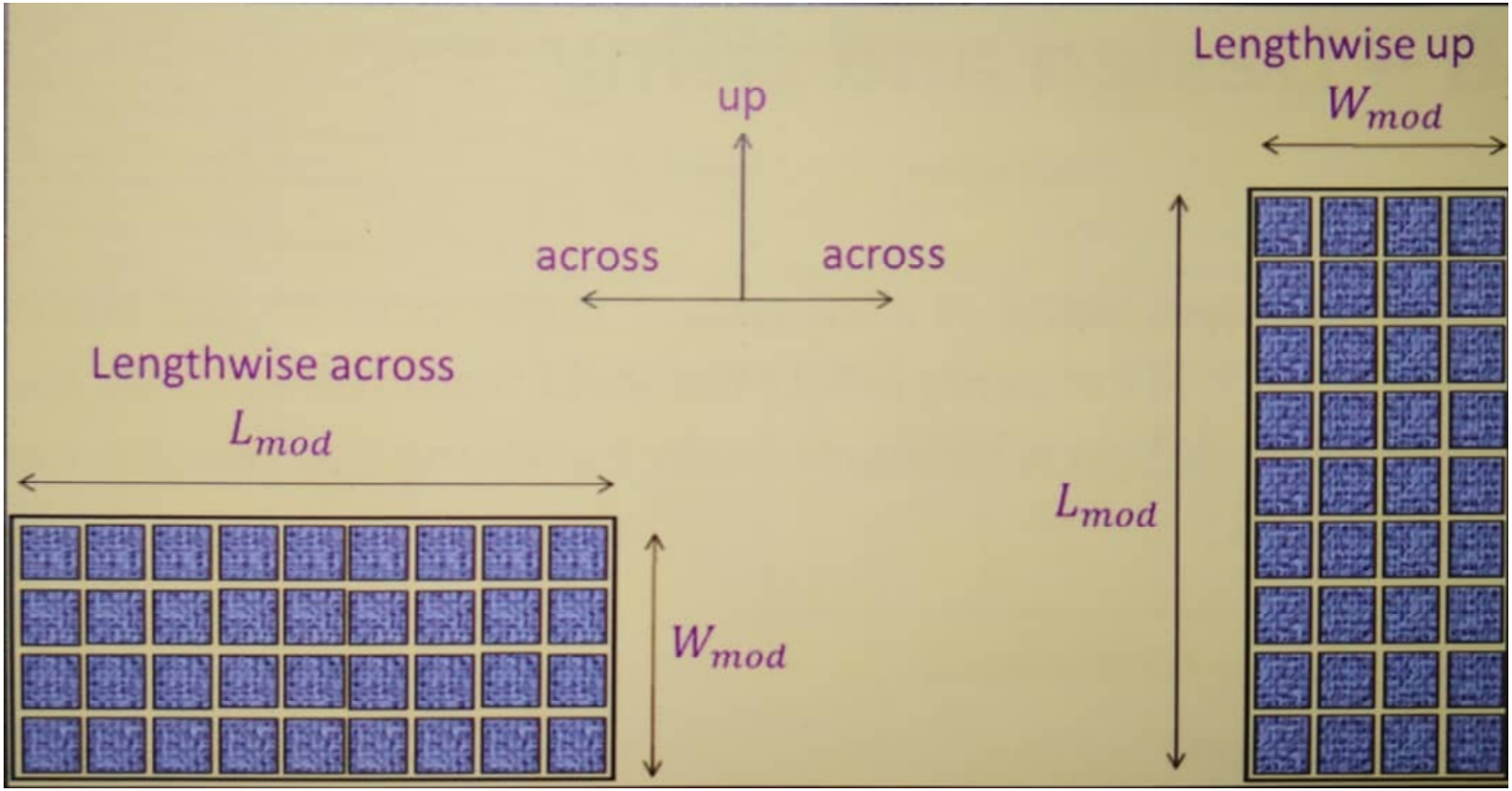
SPACE CONSTRAINT

Introduction

- The second criteria to determine the solar system capacity is space constraint.
- Here, the number of solar modules installed are determined based on the space provided at the rooftop.
- There are two possible arrangements of solar modules on the rooftop.
  - Length-wise across
  - Length-wise up

SPACE CONSTRAINT

PV Modules Arrangement



SPACE CONSTRAINT

Lengthwise Across

- Under lengthwise across arrangement, the maximum number of solar modules can be installed is

$$N_{\max_mod_across} = N_{across_across} \times N_{across_up}$$

where

$$N_{across_across} = \left[\frac{L_{roof}}{L_{mod} + gap}\right]_{round\ down}$$

$$N_{across_up} = \left[\frac{W_{roof}}{W_{mod} + gap}\right]_{round\ down}$$



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Budget Constraint

BUDGET CONSTRAINT

Introduction

- Budget creates a spending plan for money usage and it is crucial to ensure any project can be finished smoothly without any financial problem.
- The cost of the system will be determined by:
  - Total cost PV array power affordable
  - Maximum number of PV modules affordable

BUDGET CONSTRAINT

Calculation

- The total PV array power affordable can be calculated as:

$$P_{arr_stc} = \frac{Budget}{K_{ind}}$$

where
Budget is the money allocated for the investment (RM)
K_{ind} is cost index of PV system (RM per Wp)

- The maximum number of PV modules affordable:

$$N_{max_mod} = \left[\frac{P_{array_stc}}{P_{mod_stc}}\right]_{round\ down}$$

where
P_{array_stc} is the maximum power of PV array at STC (Wp)
P_{mod_stc} is maximum power of PV module at STC

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Sizing of PV Array to Inverter

SIZING OF PV ARRAY TO INVERTER

Introduction

- The sizing of PV array to inverter is determined by :
  - Range of numbers of modules
  - Extreme limits of voltage
  - String modules
  - Parallel strings
  - Array configuration

SIZING OF PV ARRAY TO INVERTER

Extreme Limits of Voltage

- The maximum voltage produced by PV module:

$$V_{max} = V_{oc_stc} \times \left\{1 + \left[\left(\frac{\beta_{voltage}}{100}\right) \times \Delta T_{\min_stc}\right]\right\}$$

where

V_{max} is the maximum voltage output from PV module (V)

V_{oc_stc} is open circuit voltage at STC (V)

$\beta_{voltage}$ is temperature coefficient of voltage (%/°C)

$\Delta T_{\min_stc}$ is difference between minimum operating cell temperature and STC temperature (°C)

SIZING OF PV ARRAY TO INVERTER

Extreme Limits of Voltage

- The minimum voltage produced by PV module:

$$V_{min} = V_{mp_stc} \times \left\{1 + \left[\left(\frac{\beta_{voltage}}{100}\right) \times \Delta T_{\max_stc}\right]\right\}$$

where

V_{min} is the minimum voltage output from PV module (V)

V_{mp_stc} is voltage at maximum power at STC (V)

$\beta_{voltage}$ is temperature coefficient of voltage (%/°C)

$\Delta T_{\max_stc}$ is difference between maximum operating cell temperature and STC temperature (°C)

SIZING OF PV ARRAY TO INVERTER

String Modules

- The range of numbers of PV modules in a string:

$$N_s = [N_{s_min}]..[N_{s_max}]$$

$$N_{s_min} = \left[\frac{V_{min_inv_win} \times k_3}{V_{min} \times k_4}\right]_{round\ up}$$

$$N_{s_max} = \left[\frac{V_{max_inv} \times k_5}{V_{max}}\right]_{round\ down}$$

where

N_s is range of number of PV modules in a string
N_{s_min} is minimum number of PV modules per string
N_{s_max} is maximum number of PV modules per string
$V_{min_inv_win}$ is minimum input voltage window of inverter
V_{max_inv} is maximum input voltage of inverter
k_3 and k_4 are safety margin
k_5 is cable loss

SIZING OF PV ARRAY TO INVERTER

Parallel Strings

- For poly, mono and HIT panels (Not for thin film):

$$N_{p,max} = \left[\frac{I_{dc_inv}}{I_{sc_stc} \times 1.25}\right]_{round\ down}$$

where

N_{p_max} is maximum number of strings in parallel

I_{dc_inv} is maximum input current to the inverter

I_{sc_stc} is short-circuit current of string at STC

SIZING OF PV ARRAY TO INVERTER

Parallel Strings

- The range of numbers of PV modules in a parallel:

$$N_p = \left[\frac{N_{max}}{N_{s_max}}\right]_{round\ down}$$

Or

$$N_p = \left[\frac{N_{max}}{N_x}\right]_{round\ down}$$

where

$$N_{s_min} \leq N_x \leq N_{s_max}$$

$$1 \leq N_p \leq N_{p_max}$$

SIZING OF PV ARRAY TO INVERTER

Array Configuration

- The total array configuration:

$$N_t = N_p \times N_s$$

where

N_t is total number of PV modules in the array.



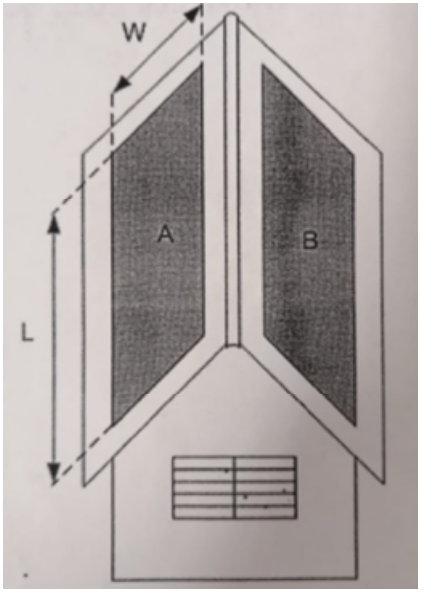
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EXAMPLES

Example 1

Which type of the following PV module could generate the highest energy based on the given roof space area (L=11m and W=6m)? How many PV modules? What is the total array power? Given gap = 10mm.

- Advent 210
- Aleo S-03 150W
- HIP-210NKHE1
- BP Solar BP3220N
- Mitsubishi PV-TJ210GA6



(HIP-210NKHE1, 84 modules, 17640 Wp)

Example 2

The maximum budget allocation is RM180,000. How many PV modules is affordable by the client? Given the cost index of PV system is RM10/Wp.

(85 modules)

Example 3

Assuming maximum efficiency of inverter is 98% and power loss in DC cable is 5%, how many PV modules are required to satisfy the energy requirement if the client requires not more than 90% of his total annual energy consumption to be supplied by PV system? Given daily PSH is 5 hour, dirt de-rate power is 3% and average temperature is 33°C.

Monthly energy consumption (kWh)

1	2	3	4	5	6	7	8	9	10	11	12
2400	2300	2200	2300	2500	NA	2300	2400	2300	2200	2250	2300

(84 modules)

Example 4

Determine the optimum PV array configuration. Given the following inverters:

- Solarmax 30c
- Fronius IG 15
- Kaco Powador 16.0 TR3
- Solarmax 20c
- Fronius IG Plus 100V-2

(4 X 10)



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QUESTIONS?